

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (BTech-MTech) 4th Semester(Mid) Examination, February, 2026
Theory of Computation (CS2204)

Full Marks: 30

Time: 2 Hour

Answer any three.

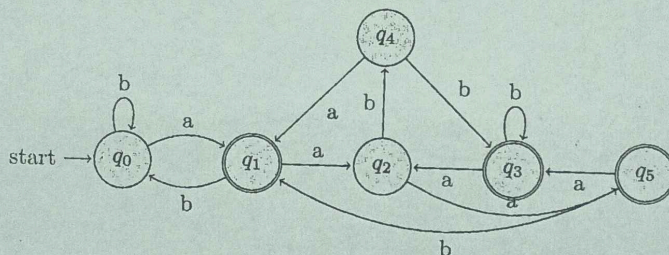
1. (a) Construct a DFA for the following Language and check whether $w = 011011$ is a valid string or not.
 $L(G) = \{w | w \in \{0, 1\}^* \text{ and } w \text{ starts with 0 and has odd lengths or it starts with 1 and has even length}\}$
 (b) Construct a DFA that accepts the set of all binary strings that, interpreted as the binary representation of an unsigned decimal integer, is divisible by 5.

[4 + 6 = 10]

2. (a) Prove that the set of Regular languages is closed under complement and intersection. Also Give examples. (i.e., for regular languages L_1 and L_2 , $\bar{L}_1$ is Regular Language, $L_1 \cap L_2$ is also regular.
 (b) Write the regular Expression for the even numbers of a's and even numbers of b's of a string w over $\{a, b\}^*$. Draw the transition diagram for the DFA that accepts the same language.
 (c) State pumping lemma. Prove that the language $L = \{ww^R : w \in \Sigma^*\}$ is not regular. $\Sigma = \{a, b\}$

[(2 + 2) + (1 + 2) + 3 = 10]

3. (a) Consider the ϵ -NFA as shown in the following figure and find a minimum state DFA that accepts the same language.



- (b) Consider the following regular Expression and draw the transition diagram accepting the same language.
 - i. $(101 + 101)^*10$
 - ii. $10 + 110^* + 01^*01$

[6 + (2 + 2) = 10]

4. (a) Proof that if L_1 and L_2 are regular languages, then $L_1 \cup L_2$, $L_1 L_2$ and L_1^* are also regular languages.
 (b) Find the context-free grammars that generate the following languages:

$$L = \{a^n b^i c^n | n \geq 1, i \geq 0\}$$

[6 + 4 = 10]